

ESP32 and R307S Fingerprint Sensor Interfacing Guide

Introduction

This document explains how to interface the R307S fingerprint sensor with an ESP32 DevKit board. The R307S communicates using UART serial communication and can be used for biometric authentication, attendance systems, access control, and smart lock applications.

Required Components

- ESP32 DevKit
- R307S Fingerprint Sensor
- Breadboard
- Jumper Wires
- Green LED
- Red LED
- 220 Ohm Resistors

R307S Pin Description

VCC - 5V Power Supply
GND - Ground
TX - UART Data Output
RX - UART Data Input
TOUCH - Finger Detection Output
3.3V - Auxiliary/Internal Voltage Pin

ESP32 Connections

R307S VCC -> ESP32 VIN (5V)
R307S GND -> ESP32 GND
R307S TX -> ESP32 GPIO16 (RX2)
R307S RX -> ESP32 GPIO17 (TX2)

Communication Method

The R307S uses TTL UART communication. The ESP32 communicates with the sensor through UART2 configured at 57600 baud rate. The sensor performs fingerprint processing internally and returns fingerprint IDs and status information to the ESP32.

Enrollment Process

1. Enter an ID number.
2. Place a finger on the sensor.
3. Remove the finger.
4. Place the same finger again.
5. The sensor creates a fingerprint template.
6. The template is stored in internal flash memory.

Authentication Process

1. User places a finger on the sensor.
2. Sensor captures the fingerprint image.
3. Features are extracted.
4. Stored templates are searched.
5. Matching ID is returned.
6. ESP32 performs the desired action.

LED Authentication Project

Authorized Fingerprint -> Green LED ON

Unauthorized Fingerprint -> Red LED ON

This demonstrates biometric authentication using ESP32 and R307S.

ESP32 UART Initialization Example

```
HardwareSerial mySerial(2);  
mySerial.begin(57600, SERIAL_8N1, 16, 17);
```

Project Flow

Power ON -> Wait for Finger -> Capture Fingerprint -> Search Database -> Match Found? -> Green LED : Red LED

Applications

Attendance Systems
Smart Door Locks
Access Control Systems
Biometric Authentication
Embedded Security Systems

Skills Demonstrated

Embedded C/C++
UART Communication
ESP32 Programming
Sensor Interfacing
Biometric Authentication
Embedded System Design